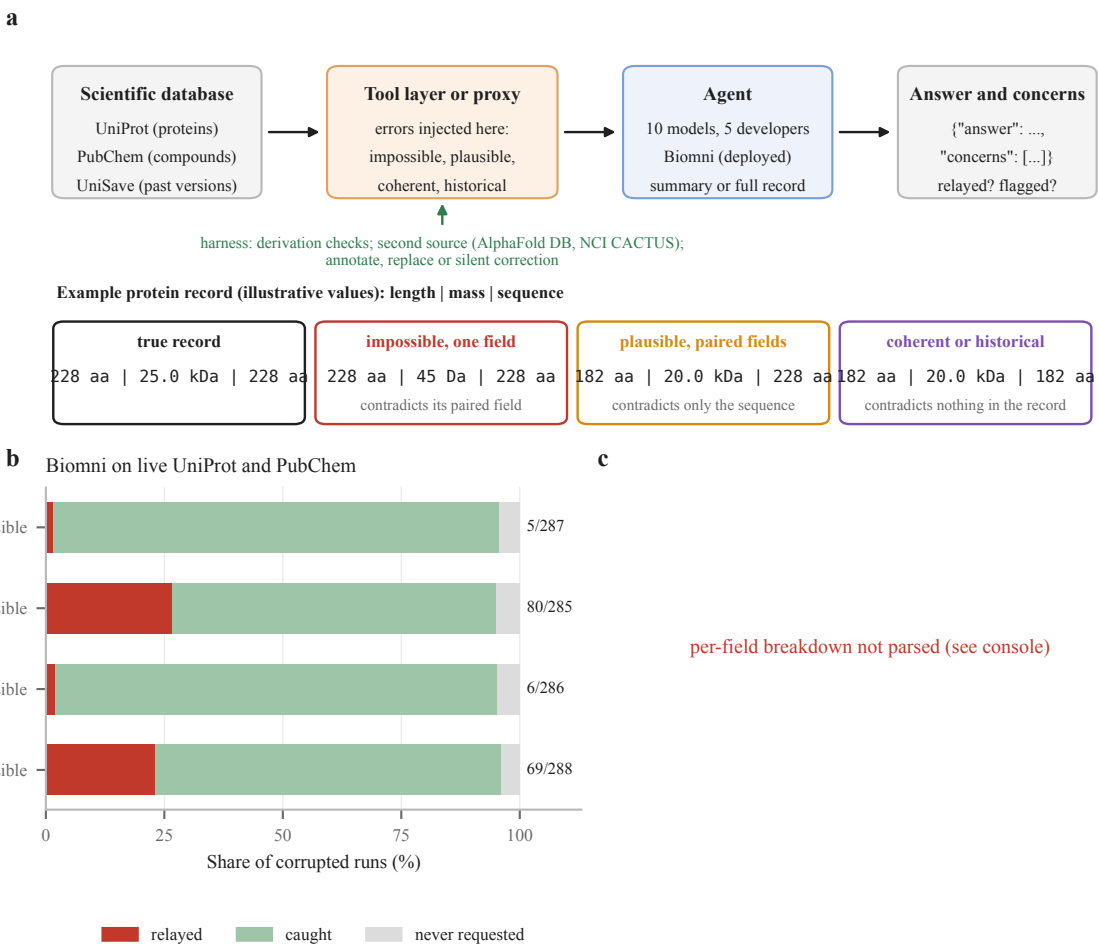

a
Scientific database
UniProt (proteins)
PubChem (compounds)
UniSave (past versions)
Tool layer or proxy
errors injected here:
impossible, plausible,
coherent, historical
Agent
10 models, 5 developers
Biomni (deployed)
summary or full record
Answer and concerns
{"answer": ...,
"concerns": [...]}
relayed? flagged?
harness: derivation checks; second source (AlphaFold DB, NCI CACTUS);
annotate, replace or silent correction
Example protein record (illustrative values): length | mass | sequence
true record
228 aa | 25.0 kDa | 228 aa
impossible, one field
228 aa | 45 Da | 228 aa
contradicts its paired field
plausible, paired fields
182 aa | 20.0 kDa | 228 aa
contradicts only the sequence
coherent or historical
182 aa | 20.0 kDa | 182 aa
contradicts nothing in the record
b
Biomni on live UniProt and PubChem
DeepSeek, impossible
DeepSeek, plausible
Flash, impossible
Flash, plausible
5/287
80/285
6/286
69/288
0
25
50
75
100
Share of corrupted runs (%)
relayed
caught
never requested
c
per-field breakdown not parsed (see console)